Appendix 1 the full search strategy of each database

Database	Search strategy
PubMed	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. (lung cancer or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge"[Mesh] OR "Health Knowledge, Attitudes, Practice"[Mesh]) OR "Awareness"[Mesh]) OR ("Attitude"[Mesh] OR "Attitude to Health"[Mesh])) OR "Behavior"[Mesh]) OR "Patient Compliance"[Mesh]) OR "Perception"[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitudes or practice or belief or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

EMBASE	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab/Keyword] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab/Keyword] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
PsycINFO	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh]

	10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
Ovid (Journals @ Ovid Full Text and Ovid MEDLINE)	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

CINAHL	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab/MM] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[MM] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[MM] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

Cochrane Library	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. (lung cancer or NSLC or NSCLC or SLC or SCLC) [ti/ab/kw] 3#. (lung neoplasm*)[ti/ab/kw] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT) [ti/ab/kw] 6# scan* or screen*[ti/ab/kw] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab/kw] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitudes or practice or belief or adherence or compliance or perception or perspective)[ti/ab/kw] 13#. 11 or 12 14#. 4 and 10 and 13
Chinese Biomedical Database (CBM)	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[ti/ab]

	10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/ab] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
Chinese National Knowledge Infrastructure (CNKI)	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/kw/ab] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/kw/ab] 3#. (lung neoplasm*)[ti/kw/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/kw/ab] 6# scan* or screen*[ti/kw/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/kw/ab] 9#. "Early Detection of Cancer"[ti/kw/ab] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/kw/ab] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/kw/ab] 13#. 11 or 12 14#. 4 and 10 and 13

Chinese Wanfang Database	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/kw] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/kw] 3#. (lung neoplasm*)[ti/kw] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/kw] 6# scan* or screen*[ti/kw] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/kw] 9#. "Early Detection of Cancer"[ti/kw] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/kw] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/kw] 13#. 11 or 12 14#. 4 and 10 and 13

Note: MM: MeSH; ti: title; ab: abstract; kw: keyword

Appendix 2 Structured and standardized data extraction forms

Data items	Description
Reference ID number	ID number in citation management software
Author(s), year of publication	First author, article year
Geographic location	In which country/city was the study conducted
Study setting	Where did the study take place (including context).
Study design	The study design as defined by authors.
Research purpose/ focus	The aims of the study as defined by the author.
Population	Number, types and characteristics of participants of the study, including screening time and risk status. (sampling methods, sampling numbers, age (range & mean), gender(%), smoking status, screening individuals or healthcare providers, screening time and risk status(n(%))
Outcomes	How the authors measured outcomes and the main results. Intention to screen: n(%)
Barriers	Factors that may have reduced uptake of LCS at the level of individuals and healthcare providers.
Facilitators	Factors that may have enabled uptake of LCS at the level of individuals and healthcare providers.
Theoretical basis	If applicable, theories and frameworks described in the study for the categorization of barriers and facilitators.
Study limitations	Authors' reported gaps and limitations of the study.
Study implications	Authors' reported future research direction.

Appendix 3 Data synthesize forms

Data items	**Social-ecological model**	**Subgroup analysis**			**Outcome description**			
Barriers /facilitators	Including individual, interpersonal, health system, and socio/environmental factors	Screening time	Risk status	Individuals	Identified in a quantitative study (# studies & results)	Identified in a qualitative study (# studies & results)	Identified in a Mixed methods study (# studies& results)	Total # studies in which identified

Note: screening time (initial vs ongoing), risk status (high-risk, moderate-risk, vs low-risk), and individuals (participants vs healthcare providers).